Daily arXiv Digest - Dynamical Systems

ARXIV math.DS · 2026-09-07

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The live math.DS feed contains 18 unseen announcements: six high priority, eight related, and four low priority. The closest matches concern subshifts of finite type on groups, Hausdorff measure and dimension, self-similar measures, transfer operators, and measure and orbit equivalence. Related entries include topological classification, centralizer rigidity, complex dynamics, and geometric or operator-algebraic questions about groups. Every digest is based only on the supplied title and abstract; claims of proofs and constructions are attributed to the authors, and numerical evidence is identified separately.

Only new and newly cross-listed papers are included; replacement-only submissions are excluded. Mathematical claims are constrained to the title and abstract.

HIGH PRIORITY

1. Medvedev degrees of SFTs on cocompact Fuchsian groups

Sebastián Barbieri, Nicanor Carrasco-Vargas, Paul Toussaint
arXiv:2609.04353 · new · abstract · PDF

Relevance	Subshifts of finite type on cocompact Fuchsian groups directly match symbolic dynamics and general group actions.
TL;DR	The authors identify the Medvedev degrees attained by subshifts of finite type on every cocompact Fuchsian group. These are exactly the Π^0_1 degrees.
Problem	Determine the class of Medvedev degrees realized by subshifts of finite type on a cocompact Fuchsian group.
Main result	For any cocompact Fuchsian group, the authors prove that the class of Medvedev degrees attained by subshifts of finite type is exactly the class of Π^0_1 degrees.
Methods / framework	Construction of a rigid hierarchical structure in a graph model of the hyperbolic plane · A representation in $\mathrm{PSL}_2(\mathbb{R})$ whose matrices have computable real coefficients
Context	Symbolic dynamics on groups, Medvedev degrees, and computability in hyperbolic geometry.
Prerequisites	Subshifts of finite type · Cocompact Fuchsian groups · Medvedev degrees
Keywords	subshifts of finite type · Fuchsian groups · Medvedev degrees · computability

Original abstract

We prove that for any cocompact Fuchsian group the class of Medvedev degrees attained by subshifts of finite type is the class of Π^0_1 *degrees. This result relies on the construction of a rigid hierarchical structure in a graph model of the hyperbolic plane, and on the fact that every such group admits a representation in* $\mathrm{PSL}_2(\mathbb{R})$ *such that the matrices in the image have coefficients that are computable real numbers.*

2. On the Hausdorff measure of projections of self-similar sets

Chong-Wei Liang
arXiv:2609.04684 · new · abstract · PDF

Relevance	Hausdorff measure of projections of self-similar Cantor sets directly matches fractal geometry and dimension theory.
TL;DR	The authors obtain vanishing Hausdorff measure for almost every direction within a positive-measure set of projection directions for certain four-corner Cantor sets. The contraction ratio is strictly above the polynomial-defined threshold in the abstract and at most one sixth.
Problem	Determine when projections of a four-corner Cantor set have positive Hausdorff measure at the dimension of the original set for almost every direction.
Main result	For contraction ratios strictly greater than the threshold approximately 0.155124983896014 and at most one sixth, the authors exhibit a set of directions of positive Lebesgue measure on which almost every projection has zero Hausdorff measure at the dimension of the original Cantor set. The exact threshold is the unique root in the interval specified for the polynomial in the abstract. The assertion concerns almost every direction within that set, not almost every direction overall.
Methods / framework	Not specified in the abstract.
Context	Hausdorff measure and projections of self-similar sets; the abstract places the result within a previously open range of contraction ratios.
Prerequisites	Four-corner Cantor sets · Hausdorff measure and dimension · Projections · Lebesgue measure
Keywords	Hausdorff measure · projections · self-similar sets · Cantor sets

Original abstract

For $0 < d \le 1/4$, let $C(d)$ be the four-corner Cantor set with Hausdorff dimension $s_d = \log 4/\log(1/d)$. Peres, Simon, and Solomyak, as well as Mattila, asked for which d the measure $\text{\Hau}^{s_d}(p_\theta(C(d)))$ is positive for almost every direction θ. It was open for the range $1/9 \le d \le 1/6$. In this paper, we show that, for $\delta < d \le 1/6$, there is a set $\text{\IP}(d)$ of positive Lebesgue measure such that for almost every $\theta \in \text{\IP}(d)$, $\text{\Hau}^{s_d}(p_\theta(C(d))) = 0$, where $\delta = 0.155124983896014\ldots$ is the unique zero in $(1/9, 1/6)$ of the polynomial $P(d) = 1 - 7d + 3d^2 + 4d^3 - 2d^4 - d^5$. [Source notation: the supplied abstract does not define the macros Hau, IP; their literal command names are retained.]

3. Cone upgrade for effective equidistribution and weighted Khintchine–Schmidt theorem on fractals

Gaurav Aggarwal, Alexander Gorodnik
arXiv:2609.04953 · new · abstract · PDF

Relevance	The main results concern self-similar measures and Diophantine approximation on fractals, with an explicit homogeneous-dynamics connection.
TL;DR	The authors prove weighted Khintchine and Schmidt analogues for products of self-similar measures on the real line whose defining systems share a common contraction ratio. Their cone upgrade derives effective double equidistribution, uniform over weights in a fixed cone, from effective equidistribution for one weight.
Problem	Establish weighted counting theorems on products of self-similar measures using effective homogeneous-dynamics equidistribution.

Main result	Under the common-contraction-ratio assumption, the authors prove fractal analogues of the weighted Khintchine and Schmidt theorems. They obtain effective double equidistribution for translates of the product measures, uniformly over diagonal weights ranging in a fixed cone in the positive Weyl chamber, from a single-weight effective equidistribution statement. The abstract also states a partial effective form of a theorem of Khalil, Luethi and Weiss without specifying its precise formulation.
Methods / framework	Translation of a counting problem into homogeneous dynamics · Cone upgrade from effective equidistribution for a single weight · Effective double equidistribution
Context	Self-similar measures, weighted Diophantine approximation, and homogeneous dynamics. The abstract explicitly describes building on prior work treating the unweighted case.
Prerequisites	Self-similar measures · Iterated function systems · Homogeneous dynamics · Effective equidistribution
Keywords	fractals · self-similar measures · weighted approximation · equidistribution

Original abstract

We prove a fractal analogue of the weighted theorems of Khintchine and Schmidt for products of self-similar measures on the real line whose defining iterated function systems share a common contraction ratio. The proof translates the counting problem into homogeneous dynamics and builds on recent work of Bénard, He and Zhang, who treated the unweighted case. Our main new ingredient is a cone upgrade: an effective double equidistribution theorem for translates of product self-similar measures, uniform over diagonal elements whose weights range over a fixed cone in the positive Weyl chamber, deduced from effective equidistribution for a single weight. This yields a partial effective form of a theorem of Khalil, Luethi and Weiss.

4. Relative (τ), Expanders, and Decay of Correlations for certain Expanding Maps

Rhiannon Dougall
arXiv:2609.05271 · new · abstract · PDF

Relevance	Subshifts of finite type, transfer operators, group quotients, and mixing directly connect to symbolic dynamics and group-related dynamical questions.
TL;DR	The paper investigates whether expansion of group quotients leads to good mixing for towers of finite-sheeted covers of certain expanding systems. The stated framework includes subshifts of finite type and expanding interval maps, using transfer operators and connections with KMS states.
Problem	Investigate mixing phenomena associated with expander families in finite-sheeted covers of expanding dynamical systems beyond the motivating homogeneous example.
Main result	The abstract states that the investigation uses transfer-operator machinery applicable to subshifts of finite type and expanding interval maps, with connections to KMS states of Cuntz-Krieger algebras. A precise new decay-of-correlations theorem and its hypotheses are not specified in the abstract. The motivating uniform exponential-mixing statement for geodesic flows must not be read as a newly asserted conclusion for all systems considered here.
Methods / framework	Transfer operators · Connections with KMS states of Cuntz-Krieger algebras
Context	Expander families, relative tau, mixing, symbolic dynamics, and expanding maps.
Prerequisites	Subshifts of finite type · Expanding maps · Expander graphs · Transfer operators · KMS states
Keywords	subshifts of finite type · transfer operators · expanders · decay of correlations

Original abstract

Relative (τ) is equivalent to a statement that the sequence of Cayley graphs associated to group quotients $\Gamma_q = \Gamma/N_q$, $q \in \mathbb{N}$, form an expander family. There is a philosophy that expander graphs give rise to good mixing; for instance, one has exponential mixing for the geodesic flow uniformly the along a family of congruence covers of the modular surface, stemming from the symmetry in the $\mathrm{SL}(2,\mathbb{R})$ action and the uniform spectral gap for the Laplacian. Do we see similar phenomena in less structured settings? We investigate this question for a tower of finite sheeted covers of certain expanding dynamical systems. We use transfer operator machinery that applies in particular in the cases of subshifts of finite type and expanding interval maps. We make fruitful connections with KMS states of Cuntz–Krieger algebras.

5. Distribution of points near the origin in the d-dimensional Lagrange spectrum

Benjamin Ward
arXiv:2609.05311 · cross · abstract · PDF

Relevance	Hausdorff and box-dimension estimates for Diophantine approximation sets directly match the dimension-theoretic interests.
TL;DR	In simultaneous Diophantine approximation in dimension at least two, the authors obtain Hausdorff-dimension results for sets with best-approximation constants in narrow intervals near zero. They also give a box-dimension lower bound for the Lagrange spectrum and an uncountability result for square systems of linear forms.
Problem	Study the size of sets with prescribed small best-approximation constants and the distribution of values in higher-dimensional Lagrange spectra.
Main result	The authors prove positive Hausdorff dimension for points whose best-approximation constants lie in the interval specified in the abstract, with some positive interval-width constant. For a slightly larger set, the dimension approaches full dimension as the lower endpoint tends to zero; the larger set is not specified. They bound the box dimension of the spectrum below by one minus the reciprocal of one more than the ambient dimension. They also prove uncountability of the naturally defined spectrum for square systems of linear forms.
Methods / framework	A framework inspired by Schmidt's games · Application of the Simplex lemma near rational points · An elementary observation for the square-matrix uncountability result, not detailed in the abstract
Context	Simultaneous Diophantine approximation, Lagrange spectra, Hausdorff dimension, and box dimension.
Prerequisites	Diophantine approximation · Lagrange spectra · Hausdorff and box dimension · Schmidt's games
Keywords	Hausdorff dimension · box dimension · Lagrange spectrum · Diophantine approximation

Original abstract

We develop a new framework, inspired by Schmidt's games, to study the Lagrange spectrum for simultaneous Diophantine approximation in dimension $d \geq 2$. We show the Hausdorff dimension of the set of points in $\mathbb{R}^d$ whose best approximation constant lies in $[\varepsilon, \varepsilon(1+\delta\varepsilon^d)]$, for some constant $\delta > 0$, is positive, and for a slightly larger set approaches full dimension as $\varepsilon \to 0$. We also show that the box dimension of the d-dimensional Lagrange spectrum is bounded from below by $1 - \frac{1}{d+1}$. The proof combines a novel application of the Simplex lemma near rational points with the game-theoretic framework. As an additional result we use an elementary observation to show that the naturally defined Lagrange spectrum for systems of linear forms is uncountable in the case of square matrices.

6. Graph products and measure equivalence: classification, rigidity, and quantitative aspects

Amandine Escalier, Camille Horbez
arXiv:2401.04635 · replace-cross · abstract · PDF

Relevance	Probability measure-preserving group actions, measure equivalence, and quantitative orbit equivalence are central to this paper and match the general-group-action interests.
TL;DR	The authors classify graph products up to measure equivalence under stated restrictions on the defining graphs and vertex groups, with a quantitative version tracking cocycle integrability. They also obtain orbit-equivalence rigidity under extra ergodicity assumptions and measure-equivalence rigidity results among torsion-free groups.
Problem	Classify graph products of groups in measured group theory and determine rigidity and quantitative orbit-equivalence consequences.
Main result	The first classification applies to graph products of countably infinite groups over finite simple graphs with no transvection and no partial conjugation. A quantitative version tracks cocycle integrability and yields an inverse-problem result for a large family of right-angled Artin groups. Orbit-equivalence rigidity for probability measure-preserving actions requires additional ergodicity assumptions not specified here. The authors give a sufficient graph-and-vertex-group condition for measure-equivalence rigidity among torsion-free groups and construct a group rigid in that sense but not in quasi-isometry. The complete classification criteria are not specified in the abstract.
Methods / framework	Quantitative tracking of cocycle integrability · Construction using variations of Higman groups as vertex groups
Context	Measured group theory, probability measure-preserving actions, orbit equivalence, and associated von Neumann algebras. Additional applications to commensurability, isomorphism, and automorphism groups are stated without their detailed formulations.
Prerequisites	Graph products of groups · Measure equivalence · Orbit equivalence · Probability measure-preserving actions · Cocycles
Keywords	group actions · measure equivalence · quantitative orbit equivalence · rigidity

Original abstract

We study graph products of groups from the viewpoint of measured group theory. We first establish a full measure equivalence classification of graph products of countably infinite groups over finite simple graphs with no transvection and no partial conjugation. This finds applications to their classification up to commensurability, and up to isomorphism, and to the study of their automorphism groups. We also derive structural properties of von Neumann algebras associated to probability measure-preserving actions of graph products. Variations of the measure equivalence classification statement are given with fewer assumptions on the defining graphs. We also provide a quantified version of our measure equivalence classification theorem, that keeps track of the integrability of associated cocycles. As an application, we solve an inverse problem in quantitative orbit equivalence for a large family of right-angled Artin groups. We then establish several rigidity theorems. First, in the spirit of work of Monod-Shalom, we achieve rigidity in orbit equivalence for probability measure-preserving actions of graph products, upon imposing extra ergodicity assumptions. Second, we establish a sufficient condition on the defining graph and on the vertex groups ensuring that a graph product G is rigid in measure equivalence among torsion-free groups (in the sense that every torsion-free countable group H which is measure equivalent to G, is in fact isomorphic to G). Using variations over the Higman groups as the vertex groups, we construct the first example of a group which is rigid in measure equivalence, but not in quasi-isometry, among torsion-free groups.

RELATED / POSSIBLY INTERESTING

1. Loss of memory in the periodic Ehrenfest model with small polyhedral scatterers

Christopher Lutsko, Jens Marklof
arXiv:2609.04371 · new · abstract · PDF

Relevance	The loss-of-memory result concerns statistical behavior in a periodic billiard model. Its connection to the core entropy and symbolic interests is not stated; it is conservatively retained as potentially related ergodic dynamics.
TL;DR	The authors determine a Boltzmann-Grad limit for the periodic Ehrenfest wind-tree model with polyhedral scatterers. For a generic polyhedron in dimension at least three, they obtain a loss-of-memory effect leading to a Markovian transport process on an extended state space.
Problem	Determine the limiting transport dynamics of the periodic Ehrenfest model with small polyhedral scatterers in the Boltzmann-Grad limit.
Main result	The central stated result is loss of memory in dimension at least three for a generic choice of polyhedron. The authors derive a Markovian limiting process on an extended state space and a generalised linear Boltzmann equation for particle-density evolution; the meaning of genericity is not specified in the abstract.
Methods / framework	Not specified in the abstract.
Context	Periodic billiard dynamics, transport limits, and loss of memory.
Prerequisites	Ehrenfest wind-tree model · Boltzmann-Grad limits · Markov processes
Keywords	periodic scatterers · loss of memory · Boltzmann-Grad limit · transport

Original abstract

We determine the Boltzmann–Grad limit of the Ehrenfest “wind-tree” model with a periodic configuration of polyhedral scatterers. The central result is a loss-of-memory effect in dimension $d \geq 3$ for a generic choice of polyhedron. This yields a Markovian limiting transport process on an extended state space and a generalised linear Boltzmann equation for the time evolution of the particle density.

2. Generalized upper principal part of real planar polynomial vector fields

Thaís Maria Dalbelo, Regilene Oliveira, Otavio Henrique Perez
arXiv:2609.04413 · new · abstract · PDF

Relevance	Although centered on planar polynomial vector fields, the result concerns topological classification and equivalence near infinity.
TL;DR	The authors define a minimal generalized upper principal part by deleting selected monomials. On a stated open dense subset of vector fields with Newton degenerate upper principal part, this truncation is topologically equivalent to the original field near infinity, subject to additional non-degeneracy assumptions.
Problem	Classify the dynamics of real planar polynomial vector fields near infinity using a reduced principal part defined through Newton polyhedra.

Main result	The authors prove the existence of an open dense subset within the set of polynomial vector fields having Newton degenerate upper principal part. For fields in that subset, the original field and its minimal generalized upper principal part are topologically equivalent near infinity, provided the latter satisfies additional non-degeneracy assumptions. Those additional assumptions are not specified in the abstract.
Methods / framework	Newton polyhedra · Toric compactification · Normal Form Theorem
Context	Topological classification near infinity for planar polynomial dynamics.
Prerequisites	Planar polynomial vector fields · Newton polyhedra · Topological equivalence
Keywords	topological classification · principal part · dynamics near infinity · toric compactification

Original abstract

The goal of this paper is to generalize recent results about the topological classification of the dynamics of a real planar polynomial vector field near infinity. Given a polynomial vector field X, using Newton polyhedra one can define its generalized upper principal part X_Γ^U. By dropping some monomials of X_Γ^U, we define its minimal generalized upper principal part X_G^U. We prove that there exist an open and dense set $\widetilde{\mathfrak{U}}_1$ in the set of polynomial vector fields with Newton degenerate upper principal part satisfying the following property: if $X \in \widetilde{\mathfrak{U}}_1$, then X and X_G^U are topologically equivalent near infinity, provided that X_G^U satisfies some additional non-degeneracy assumptions. Our techniques rely on toric compactification and the Normal Form Theorem.

3. Local centralizer rigidity for a non-generic diagonal map

Zhijing Wendy Wang, Amie Wilkinson
arXiv:2609.04643 · new · abstract · PDF

Relevance	Centralizer rigidity for diagonal maps in compact homogeneous spaces connects homogeneous dynamics with symmetries of dynamical systems.
TL;DR	The authors prove centralizer rigidity for any non-trivial diagonal map in the compact homogeneous space specified in the abstract, in matrix size at least five. The perturbed volume-preserving diffeomorphism is assumed to have a regular isomorphic centralizer.
Problem	Establish local centralizer rigidity for a non-generic diagonal map on a compact homogeneous space.
Main result	The stated rigidity theorem applies to any non-trivial diagonal map in the indicated compact quotient of the real special linear group, in matrix size at least five, under the regular-isomorphic-centralizer assumption for the perturbed volume-preserving diffeomorphism. The meaning of regular isomorphic centralizer and the precise rigidity conclusion are not specified in the abstract.
Methods / framework	Not specified in the abstract.
Context	Homogeneous dynamics, volume-preserving perturbations, and centralizer rigidity.
Prerequisites	Compact homogeneous spaces · Diagonal maps · Centralizers · Volume-preserving diffeomorphisms
Keywords	centralizer rigidity · diagonal maps · homogeneous dynamics

Original abstract

In this paper, we prove centralizer rigidity for any non-trivial diagonal map in the compact homogeneous space $SL_n\mathbb{R}/\Gamma$, $n \geq 5$, under the assumption that the perturbed volume-preserving diffeomorphism has a regular isomorphic centralizer.

4. From quadratic integrals to Nijenhuis operators: two-dimensional dictionary

Dinmukhammed Akpan
arXiv:2609.05103 · cross · abstract · PDF

Relevance	The explicit correspondence concerns geodesic-flow integrability and local geometric classifications. A core entropy or symbolic connection is not stated; it is conservatively retained as potentially related smooth dynamics.
TL;DR	The authors construct an explicit correspondence between two local classifications in dimension two. It relates pseudo-Riemannian metrics with a quadratic integral of the geodesic flow to $\mathfrak{gl}$-regular Nijenhuis operators.
Problem	Relate local normal forms of pseudo-Riemannian metrics admitting a quadratic geodesic-flow integral to local normal forms of $\mathfrak{gl}$-regular Nijenhuis operators in dimension two.
Main result	An explicit dictionary between these two local classifications is constructed in dimension two. The correspondence is realized by a coordinate change built from the trace and determinant of the operator.
Methods / framework	Explicit coordinate change constructed from the operator's trace and determinant
Context	Geodesic-flow integrability, pseudo-Riemannian metrics, and Nijenhuis operators.
Prerequisites	Geodesic flows · Quadratic integrals · Pseudo-Riemannian metrics · Nijenhuis operators
Keywords	geodesic flow · quadratic integrals · normal forms · Nijenhuis operators

Original abstract

We construct an explicit dictionary between two local classifications in dimension two: normal forms of pseudo-Riemannian metrics with a quadratic integral of the geodesic flow and normal forms of $\mathfrak{gl}$-regular Nijenhuis operators. The correspondence is obtained by an explicit change of coordinates constructed from the trace and determinant of the operator.

5. Capture components of cubic Siegel polynomials

Xiaoguang Wang, Fei Yang
arXiv:2312.10929 · replace-cross · abstract · PDF

Relevance	The result gives a topological description of parameter-space components in complex dynamics under an explicit bounded-type assumption.
TL;DR	For an irrational rotation number of bounded type, the authors prove that all capture components in the specified one-parameter family of cubic Siegel polynomials are Jordan domains.
Problem	Determine the topology of capture components in the parameter space of the cubic Siegel-polynomial family displayed in the abstract.
Main result	All capture components for that family are Jordan domains, assuming the rotation number is irrational and of bounded type. The parameter ranges over the complex numbers; the assertion is restricted to the displayed family.
Methods / framework	Not specified in the abstract.
Context	Complex polynomial dynamics and the topology of parameter spaces.
Prerequisites	Cubic Siegel polynomials · Bounded-type irrational numbers · Capture components · Jordan domains

Keywords	complex dynamics · Siegel polynomials · capture components · Jordan domains

Original abstract

Let θ be an irrational number of bounded type. We prove that all capture components in the parameter space of cubic polynomials $f_a(z) = e^{2\pi i\theta} z + az^2 + z^3$, where $a \in \mathbb{C}$, are Jordan domains.

6. Affine laminations and coaffine representations

Marit Bobb, James Farre
arXiv:2404.14284 · replace-cross · abstract · PDF

Relevance	Surface-group representations and convex cocompact actions provide a group-action connection; the dimension statement concerns a parameter space rather than fractal or mean dimension.
TL;DR	The authors study convex cocompact surface-group actions with coaffine image and describe their bending data by convex projective structures and affine measured laminations. For the compatible bending-data space associated with the stated holonomy, they prove that its projectivization is a sphere of the dimension given in the abstract.
Problem	Describe the bending laminations and compatible bending-data space of the coaffine convex cocompact surface-group representations under study.
Main result	The boundary of the convex core is described as stratified, with one-dimensional strata forming a pair of bending laminations. The authors identify the bending data on each component with a convex projective structure and an affine measured lamination depending on that structure. They prove that the projectivized space of bending data compatible with the given Hitchin holonomy is a sphere of dimension $6g - 7$, in the abstract's notation.
Methods / framework	Not specified in the abstract.
Context	Convex projective geometry, surface-group actions, coaffine representations, and measured laminations.
Prerequisites	Surface groups · Convex cocompact actions · Projective structures · Hitchin holonomy · Measured laminations
Keywords	group actions · coaffine representations · bending laminations · convex projective geometry

Original abstract

We study surface subgroups of $\mathrm{SL}(4,\mathbb{R})$ acting convex cocompactly on $\mathbb{R}\mathrm{P}^3$ with image in the coaffine group. The boundary of the convex core is stratified, and the one dimensional strata form a pair of bending laminations. We show that the bending data on each component consist of a convex $\mathbb{R}\mathrm{P}^2$ structure and an affine measured lamination depending on the underlying convex projective structure on S with (Hitchin) holonomy $\rho : \pi_1 S \to \mathrm{SL}(3,\mathbb{R})$. We study the space $\mathcal{ML}^\rho(S)$ of bending data compatible with ρ and prove that its projectivization is a sphere of dimension $6g-7$.

7. ICC property (T) groups without W*-superrigidity

Shuoxing Zhou
arXiv:2608.02327 · replace-cross · abstract · PDF

Relevance	Rigidity of countable groups through their von Neumann algebras is adjacent to measured-group and group-action questions, although no action or entropy theorem is stated in the abstract.
TL;DR	The authors claim an explicit construction of two non-isomorphic countable discrete ICC groups with Kazhdan's property (T) whose group von Neumann algebras are isomorphic. They present this as a counterexample to the stated Connes rigidity conjecture; the abstract alone does not independently verify the claim.
Problem	Determine whether an ICC property (T) group is determined up to group isomorphism by its group von Neumann algebra.
Main result	The abstract asserts the existence of two explicit countable discrete groups, both ICC and with Kazhdan's property (T), which are non-isomorphic but have isomorphic group von Neumann algebras. It identifies this asserted construction as a counterexample to Connes' rigidity conjecture for such groups. The groups and the proof are not specified in the abstract.
Methods / framework	Not specified in the abstract.
Context	Countable discrete groups, Kazhdan's property (T), and von Neumann algebra rigidity. The abstract's statement about AI assistance is provenance, not evidence for the mathematical claim or a specified proof method.
Prerequisites	ICC groups · Kazhdan's property (T) · Group von Neumann algebras
Keywords	group rigidity · ICC groups · property (T) · von Neumann algebras

Original abstract

We construct two explicit countable discrete groups Γ_1 and Γ_2 that are both ICC and have Kazhdan's property (T). Although Γ_1 and Γ_2 are not isomorphic as groups, their group von Neumann algebras are isomorphic: $L(\Gamma_1) \cong L(\Gamma_2)$. This provides a counterexample to Connes' rigidity conjecture for ICC property (T) groups. This result was obtained with assistance from GPT-5.6 Sol, independently of and concurrently with work by OpenAI.

8. Random Hamiltonians II: A central limit theorem and the Hofer geometry of random walks

Adrian Dawid
arXiv:2608.07673 · replace-cross · abstract · PDF

Relevance	The geometry of Hamiltonian diffeomorphism groups and quasimorphisms gives a connection to groups of dynamical symmetries, without an explicit entropy or dimension theorem.
TL;DR	On a large class of symplectic manifolds, the authors obtain a square-root lower growth bound for the expected Hofer norm of the random walks considered. Under restriction to an abelian subgroup they also obtain a square-root upper bound, while evidence against that upper bound outside commutative subgroups is numerical.
Problem	Study the global Hofer geometry of Hamiltonian diffeomorphism groups through random walks and probabilistic growth estimates.

Main result	The abstract states the lower growth bound on a large class of symplectic manifolds, whose precise description is not given, and the upper bound for the random walk restricted to an abelian subgroup. Failure of the latter bound beyond commutative subgroups is suggested by numerical evidence rather than proved. The authors also establish Borel measurability for the previously introduced measure class in the smooth topology, measurability of stable commutator length, and a central limit theorem for Hofer-Lipschitz quasimorphisms. Full distributional hypotheses are not specified in the abstract.
Methods / framework	Random walks on Hamiltonian diffeomorphism groups · Numerical investigation of the noncommutative case
Context	Hamiltonian diffeomorphism groups, Hofer geometry, commutative subgroups, and probabilistic limit theorems.
Prerequisites	Symplectic manifolds · Hamiltonian diffeomorphisms · Hofer norm · Random walks · Quasimorphisms
Keywords	Hamiltonian diffeomorphisms · Hofer geometry · random walks · central limit theorem

Original abstract

This paper investigates the global geometry of the group of Hamiltonian diffeomorphisms $\mathrm{Ham}(M,\omega)$ *using random walks. On a large class of symplectic manifolds, we show that the expected Hofer norm of such a random walk grows at least as fast as the square root of the number of steps. Furthermore, we show that if the random walk is restricted to an abelian subgroup, then the growth rate is also bounded from above by the square root of the number of steps. We also provide some numerical evidence suggesting that this upper bound fails away from commutative subgroups. This provides, for subgroups of the group of Hamiltonian diffeomorphisms, a probabilistic version of the flatness observed in commutative finite-dimensional Lie groups. En route, we show that the class of probability measures introduced in the prequel is a class of Borel measures with respect to the* C^∞*-topology on* $\mathrm{Ham}(M,\omega)$*, show the measurability of the stable commutator length, and show a central limit theorem for Hofer-Lipschitz quasimorphisms on* $\mathrm{Ham}(M,\omega)$*.*

LOW PRIORITY / PROBABLY NOT OF INTEREST

1. **Sarymsakov-Type Semigroups With Support Dominance and Nested Mixing Cores**
Shun-Pin Hsu
arXiv:2609.05050

2. **A Complete Resolution of Forsythe's Conjecture for Restarted Conjugate Gradients**
Matthew J. Colbrook, George Stepaniants, Alex Townsend
arXiv:2609.04659

3. **Energy-Consistent Splitting and Decomposition Approaches for Coupled port-Hamiltonian ODEs**
Marius Mönch, Nicole Marheineke, Andreas Bartel, Kevin Schäfers, Michael Günther
arXiv:2609.04884

4. **Dynamic models with p parameters are identified by $2p+1$ random features**
Michael Wieck-Sosa, Cosma Rohilla Shalizi
arXiv:2607.16035